

Q1 - 25 July - Shift 1

If the absolute maximum value of the function $f(x)$
 $= (x^2 - 2x + 7) e^{(4x^3 - 12x^2 - 180x + 31)}$ in the interval $[-3, 0]$ is $f(\alpha)$, then :

- (A) $\alpha = 0$ (B) $\alpha = -3$
(C) $\alpha \in (-1, 0)$ (D) $\alpha \in (-3, -1)$

Space for your notes:

Q2 - 25 July - Shift 1

The curve $y(x) = ax^3 + bx^2 + cx + 5$ touches the x-axis at the point $P(-2, 0)$ and cuts the y-axis at the point Q, where y' is equal to 3. Then the local maximum value of $y(x)$ is :

- (A) $\frac{27}{4}$ (B) $\frac{29}{4}$ (C) $\frac{37}{4}$ (D) $\frac{9}{2}$

Space for your notes:

Q3 - 25 July - Shift 2

The sum of the maximum and minimum values of the function $f(x) = |5x - 7| + [x^2 + 2x]$ is the

interval $\left[\frac{5}{4}, 2\right]$, where $[t]$ is the greatest integer $\leq t$ is _____

Space for your notes:

Q4 - 25 July - Shift 2

Let the area enclosed by the x-axis, and the tangent and normal drawn to the curve $4x^3 - 3xy^2 + 6x^2 - 5xy - 8y^2 + 9x + 14 = 0$ at the point $(-2, 3)$ be A. Then $8A$ is equal to _____

Space for your notes:

Q5 - 26 July - Shift 1

$$\text{Let } f(x) = \begin{cases} x^3 - x^2 + 10x - 7, & x \leq 1 \\ -2x + \log_2(b^2 - 4), & x > 1 \end{cases}$$

Space for your notes:

Then the set of all values of b , for which $f(x)$ has maximum value at $x = 1$, is :

(A) $(-6, -2)$

(B) $(2, 6)$

(C) $[-6, -2) \cup (2, 6]$

(D) $[-\sqrt{6}, -2) \cup (2, \sqrt{6}]$

Q6 - 26 July - Shift 2

If the maximum value of a , for which the function

$$f_a(x) = \tan^{-1} 2x - 3ax + 7 \text{ is non-decreasing in}$$

$$\left(-\frac{\pi}{6}, \frac{\pi}{6}\right), \text{ is } \bar{a}, \text{ then } f_{\bar{a}}\left(\frac{\pi}{8}\right) \text{ is equal to}$$

(A) $8 - \frac{9\pi}{4(9 + \pi^2)}$

(B) $8 - \frac{4\pi}{9(4 + \pi^2)}$

(C) $8\left(\frac{1 + \pi^2}{9 + \pi^2}\right)$

(D) None of these

Q7 - 27 July - Shift 1

$$I = \int_{\pi/4}^{\pi/3} \left(\frac{8 \sin x - \sin 2x}{x} \right) dx. \text{ Then}$$

Space for your notes:

(A) $\frac{\pi}{2} < I < \frac{3\pi}{4}$

(B) $\frac{\pi}{5} < I < \frac{5\pi}{12}$

(C) $\frac{5\pi}{12} < I < \frac{\sqrt{2}}{3} \pi$

(D) $\frac{3\pi}{4} < I < \pi$

Q8 - 27 July - Shift 1

Questions

MathonGo

Let a function $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined as :

$$f(x) = \begin{cases} \int_0^x (5 - |t - 3|) dt, & x > 4 \\ x^2 + bx, & x \leq 4 \end{cases}$$

where $b \in \mathbb{R}$. If f is continuous at $x = 4$, then

which of the following statements is NOT true ?

(A) f is not differentiable at $x = 4$

(B) $f'(3) + f'(5) = \frac{35}{4}$

(C) f is increasing in $\left(-\infty, \frac{1}{8}\right) \cup (8, \infty)$

(D) f has a local minima at $x = \frac{1}{8}$

Space for your notes:

Q9 - 27 July - Shift 1

Let M and N be the number of points on the curve $y^5 - 9xy + 2x = 0$, where the tangents to the curve are parallel to x -axis and y -axis, respectively. Then the value of $M + N$ equals _____.

Space for your notes:

Q10 - 27 July - Shift 2

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Questions

MathonGo

A water tank has the shape of a right circular cone with axis vertical and vertex downwards. Its semi-vertical angle is $\tan^{-1} \frac{3}{4}$. Water is poured in it at a constant rate of 6 cubic meter per hour. The rate (in square meter per hour), at which the wet curved surface area of the tank is increasing, when the depth of water in the tank is 4 meters, is _____.

*Space for your notes:***Q11 - 28 July - Shift 1**

If the minimum value of $f(x) = \frac{5x^2}{2} + \frac{\alpha}{x^5}$, $x > 0$, is

Space for your notes:

14, then the value of α is equal to :

- (A) 32 (B) 64
(C) 128 (D) 256

Q12 - 28 July - Shift 1

Let $f : [0, 1] \rightarrow \mathbb{R}$ be a twice differentiable function in $(0, 1)$ such that $f(0) = 3$ and $f(1) = 5$. If the line $y = 2x + 3$ intersects the graph of f at only two distinct points in $(0, 1)$, then the least number of points $x \in (0, 1)$, at which $f''(x) = 0$, is _____.

*Space for your notes:***Q13 - 28 July - Shift 2**

The function $f(x) = xe^{x(1-x)}$, $x \in \mathbb{R}$, is

Space for your notes:

(A) increasing in $\left(-\frac{1}{2}, 1\right)$

(B) decreasing in $\left(\frac{1}{2}, 2\right)$

(C) increasing in $\left(-1, -\frac{1}{2}\right)$

(D) decreasing in $\left(-\frac{1}{2}, \frac{1}{2}\right)$

Q14 - 28 July - Shift 2

The sum of the absolute maximum and absolute minimum values of the function

$f(x) = \tan^{-1}(\sin x - \cos x)$ in the interval $[0, \pi]$ is

(A) 0

(B) $\tan^{-1}\left(\frac{1}{\sqrt{2}}\right) - \frac{\pi}{4}$

(C) $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right) - \frac{\pi}{4}$

(D) $\frac{-\pi}{12}$

Q15 - 29 July - Shift 1

Questions

MathonGo

Let $f(x) = 3^{(x^2-2)^3+4}$, $x \in \mathbb{R}$. Then which of the following statements are true ?

P : $x = 0$ is a point of local minima of f

Q : $x = \sqrt{2}$ is a point of inflection of f

R : f' is increasing for $x > \sqrt{2}$

(A) Only P and Q

(B) Only P and R

(C) Only Q and R

(D) All, P, Q and R

Space for your notes:

Q16 - 29 July - Shift 2

If the tangent to the curve $y = x^3 - x^2 + x$ at the point (a, b) is also tangent to the curve $y = 5x^2 + 2x - 25$ at the point $(2, -1)$, then $|2a + 9b|$ is equal to _____.

Space for your notes:

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Answer Key

Q1 (B)

Q2 (A)

Q3 (15)

Q4 (170)

Q5 (C)

Q6 (D)

Q7 (C)

Q8 (C)

Q9 (2)

Q10 (5)

Q11 (C)

Q12 (2)

Q13 (A)

Q14 (C)

Q15 (D)

Q16 (195)

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Q1 (B)

$$f'(x) = e^{(4x^3 - 12x^2 - 180x + 31)} (12(x^2 - 2x + 7)(x + 3)(x - 5) + 2(x - 1))$$

for $x \in [-3, 0]$

$$\Rightarrow f'(x) < 0$$

 $f(x)$ is decreasing function on $[-3, 0]$ The absolute maximum value of the function $f(x)$ is at $x = -3$

$$\Rightarrow \alpha = -3$$

Q2 (A)

$y(x) = ax^3 + bx^2 + cx + 5$ is passing through $(-2, 0)$ then $8a - 4b + 2c = 5 \dots (1)$

$y'(x) = 3ax^2 + 2bx + c$ touches x-axis at $(-2, 0)$

$$12a - 4b + c = 0 \dots (2)$$

again, for $x = 0$, $y'(x) = 3$

$$c = 3 \dots (3)$$

Solving eq. (1), (2) & (3) $a = -\frac{1}{2}$, $b = -\frac{3}{4}$

$$y'(x) = -\frac{3}{2}x^2 - \frac{3}{2}x + 3$$

 $y(x)$ has local maxima at $x = 1$

$$y(1) = \frac{27}{4}$$

Q3 (15)**(15)****Q4 (170)**

(170)**Q5 (C)**

$$f(1) = 3$$

$$\text{For } x < 1, f'(x) = 3x^2 - 2x + 10 > 0$$

 $\Rightarrow f(x)$ is increasing

$$\text{For } x > 1, f'(x) < 0$$

 $\Rightarrow$ function is decreasing.

$$\lim_{x \rightarrow 1^+} f(x) = -2 + \log_2(b^2 - 4)$$

For maximum value at $x = 1$

$$3 \geq -2 + \log_2(b^2 - 4)$$

$$32 \geq b^2 - 4 > 0$$

$$b \in [-6, -2) \cup (2, 6]$$

Q6 (D)

$$f_a(x) = \tan^{-1} 2x - 3ax + 7$$

$$f'_a(x) = \frac{2}{1+4x^2} - 3a \geq 0$$

$$a \leq \left(\frac{2}{3(1+4x^2)} \right)_{\min.} \text{ at } x = \pm \frac{\pi}{6}$$

$$a_{\max} = \bar{a} = \frac{6}{9+\pi^2}$$

$$f_a\left(\frac{\pi}{8}\right) = \tan^{-1} \frac{\pi}{4} - 3 \frac{6}{9+\pi^2} \frac{\pi}{8} + 7 = \tan^{-1} \frac{\pi}{4} - \frac{9\pi}{4(\pi^2+9)} + 7$$

Q7 (C)

Consider

$$f(x) = 8\sin x - \sin 2x$$

$$f'(x) = 8\sin x - 2\cos 2x$$

$$f''(x) = -8\sin x + 4\sin 2x$$

$$= -8\sin x (1 - \cos x)$$

$$\therefore f''(x) < 0 \quad x \in \left(\frac{\pi}{4}, \frac{\pi}{3} \right)$$

$$\therefore f'(x) \text{ is } \downarrow \text{ function}$$

$$f'\left(\frac{\pi}{3}\right) < f'(x) < f'\left(\frac{\pi}{4}\right)$$

$$5 < f'(x) < \frac{8}{\sqrt{2}}$$

$$5 < f'(x) < 4\sqrt{2}$$

$$5x < f(x) < 4\sqrt{2}x$$

$$5 < \frac{f(x)}{x} < 4\sqrt{2}$$

$$\int_{\pi/4}^{\pi/3} 5 < \int \frac{f(x)}{x} < \int_{\pi/4}^{\pi/3} 4\sqrt{2}$$

$$\int_{\pi/4}^{\pi/3} 5 < \int \frac{8 \sin x - \sin 2x}{x} < \int_{\pi/4}^{\pi/3} 4\sqrt{2}$$

$$\frac{5\pi}{12} < I < \frac{\sqrt{2}\pi}{3}$$

Q8 (C)

$$\text{Given } f(x) = \begin{cases} \int_0^x (5 - |t - 3|) dt, & x > 4 \\ x^2 + bx, & x \leq 4 \end{cases}$$

$f(x)$ is continuous at $x = 4$

$$\text{So } \lim_{x \rightarrow 4^-} f(x) = \lim_{x \rightarrow 4^+} f(x) = f(4)$$

$$\text{So } 16 + 4b = \int_0^3 (2 - t) dt + \int_3^4 (8 - t) dt$$

$$\Rightarrow 16 + 4b = 15$$

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At $x = 4$

$$\text{LHD} = 2x + b = \frac{31}{4}$$

$$\text{RHD} = 5 - |x - 3| = 4$$

$$\text{LHD} \neq \text{RHD}$$

Option (A) is true

$$\text{and } f'(3) + f'(5) = \frac{23}{4} + 3 = \frac{35}{4}$$

Option (B) is true

$$\because f(x) = x^2 - \frac{x}{4} \text{ at } x \leq 4$$

$$f'(x) = 2x - \frac{1}{4}$$

This function is not increasing.

$$\text{In the interval in } x \in \left(-\infty, \frac{1}{8}\right)$$

Option (C) is NOT TRUE.

$$\text{This function } f(x) \text{ is also local minima at } x = \frac{1}{8}$$

Q9 (2)

$$y^5 - 9xy + 2x = 0$$

$$5y^4 \frac{dy}{dx} - 9x \frac{dy}{dx} - 9y + 2 = 0$$

$$\frac{dy}{dx}(5y^4 - 9x) = 9y - 2$$

$$\frac{dy}{dx} = \frac{9y - 2}{5y^4 - 9x} = 0 \text{ (for horizontal tangent)}$$

$$y = \frac{2}{9} \Rightarrow \text{Which does not satisfy the original}$$

$$\text{equation} \Rightarrow M = 0.$$

$$\text{Now } 5y^4 - 9x = 0 \text{ (for vertical tangent)}$$

$$5y^4(9y - 2) - 9y^5 = 0$$

$$y^4[45y - 10 - 9y] = 0$$

$$y = 0 \text{ (Or) } 36y = 10$$

$$y = \frac{5}{18}$$

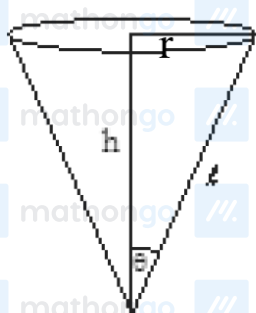
$$y = 0 \Rightarrow x = 0 \text{ \& } y = \frac{5}{18} \Rightarrow x =$$

$$(0, 0) \quad \left(x, \frac{5}{18}\right)$$

$$N = 2$$

$$M + N = 0 + 2 = 2$$

Q10 (5)



$$\tan \theta = \frac{3}{4} = \frac{r}{h}$$

$$\frac{dV}{dt} = 6$$

$$V = \frac{1}{3} \pi r^2 h = \frac{1}{3} \pi h^3 \tan^2 \theta = \frac{9\pi}{48} h^3 = \frac{3\pi}{16} h^3$$

$$\Rightarrow \frac{dV}{dt} = \frac{3\pi}{16} \cdot 3h^2 \cdot \frac{dh}{dt} = 6 \Rightarrow \left(\frac{dh}{dt} \right)_{h=4} = \frac{2}{3\pi} \text{ m/hr}$$

$$\text{Now, } S = \pi r \ell = \frac{15}{16} \pi h^2$$

$$\Rightarrow \frac{dS}{dt} = \frac{15\pi}{16} \cdot 2h \frac{dh}{dt}$$

$$\Rightarrow \left(\frac{dS}{dt} \right)_{h=4} = 5\pi \text{ m}^2/\text{hr}$$

Q11 (C)

$$\frac{x^2}{2} + \frac{x^2}{2} + \frac{x^2}{2} + \frac{x^2}{2} + \frac{x^2}{2} + \frac{\alpha}{2x^5} + \frac{\alpha}{2x^5}$$

$$\geq 7 \left(\frac{\alpha^2}{2^7} \right)^{\frac{1}{7}}$$

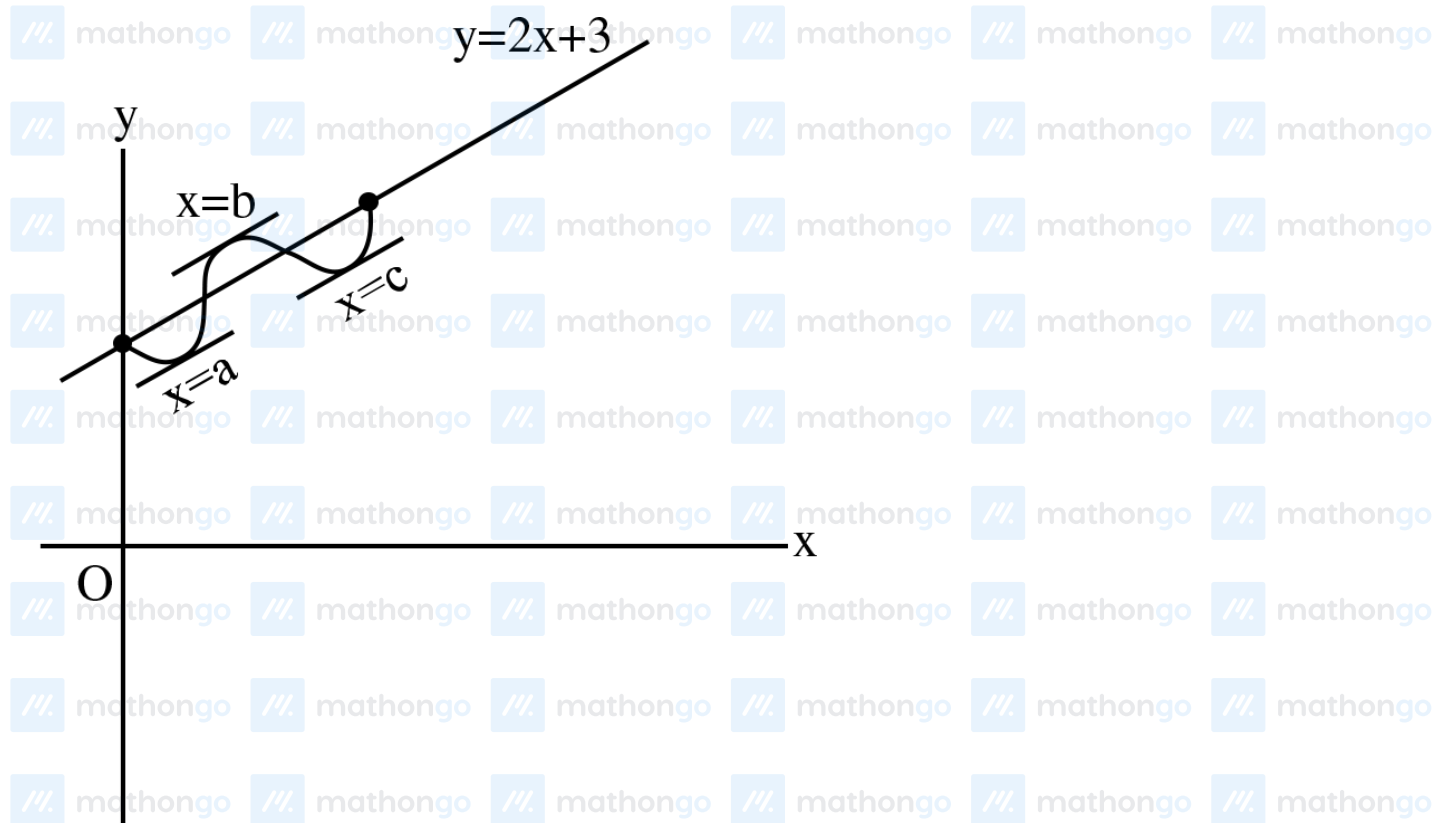
$$\frac{7 \cdot (\alpha)^{2/7}}{2} = 14$$

$$(\alpha^2)^{1/7} = 2^2$$

$$\alpha = (2^2)^{7/2} = 2^7$$

$$\alpha = 128$$

Q12 (2)



$$f'(a) = f'(b) = f'(c) = 2$$

$\Rightarrow f''(x)$ is zero

for atleast $x_1 \in (a, b)$ & $x_2 \in (b, c)$

Q13 (A)

$$f(x) = x e^{x(1-x)}$$

$$f'(x) = -e^{x(1-x)} (2x + 1) (x - 1)$$

$f(x)$ is increasing in $\left(-\frac{1}{2}, 1\right)$

Q14 (C)

$$f(x) = \tan^{-1}(\sin x - \cos x)$$

$$f'(x) = \frac{\cos x + \sin x}{(\sin x - \cos x)^2 + 1} = 0$$

$$\therefore x = \frac{3\pi}{4}$$

x	0	$\frac{3\pi}{4}$	π
f(x)	$-\frac{\pi}{4}$	$\tan^{-1}\sqrt{2}$	$\frac{\pi}{4}$

$$\therefore \left. \begin{aligned} (f(x))_{\max} &= \tan^{-1}\sqrt{2} \\ (f(x))_{\min} &= -\frac{\pi}{4} \end{aligned} \right\}$$

$$\text{sum} = \tan^{-1}\sqrt{2} - \frac{\pi}{4}$$

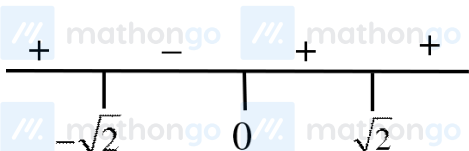
$$= \cos^{-1} \frac{1}{\sqrt{3}} - \frac{\pi}{4}$$

Q15 (D)

$$f(x) = 81.3^{(x^2-2)^3}$$

$$f'(x) = 81.3^{(x^2-2)^3} \cdot \ln 3 \cdot 3(x^2-2)^2 \cdot 2x$$

$$= (81 \times 6) 3^{(x^2-2)^3} x (x^2-2)^2 \ln 3$$



$x = 6$ is point of local min

$$f'(x) = \underbrace{(486 \cdot \ln 3)}_k \underbrace{3^{(x^2-2)^3} x (x^2-2)^2}_{g(x)}$$

$$g'(x) = 3^{(x^2-2)^3} (x^2-2)^2 + x \cdot 3^{(x^2-2)^3} \cdot 4x \cdot (x^2-2)$$

$$+ x \cdot (x^2-2)^2 \cdot 3^{(x^2-2)^3} \ln 3 \cdot 3(x^2-2)^2 \cdot 2x$$

$$= 3^{(x^2-2)^3} (x^2-2) \left[x^2-2 + 4x^2 + 6x^2 \ln 3 (x^2-2)^3 \right]$$

$$g'(x) = 3^{(x^2-2)^3} (x^2-2) \left[5x^2 - 2 + 6x^2 \ln 3 (x^2-2)^3 \right]$$

$$f''(x) = k \cdot g'(x)$$

$$f''(\sqrt{2}) = 0, f''(\sqrt{2}^+) > 0, f''(\sqrt{2}^-) < 0$$

$x = \sqrt{2}$ is point of inflection

$f''(x) > 0$ for $x > \sqrt{2}$ so $f(x)$ is increasing

Q16 (195)

$$y = 5x^2 + 2x - 25$$

$$P(2, -1)$$

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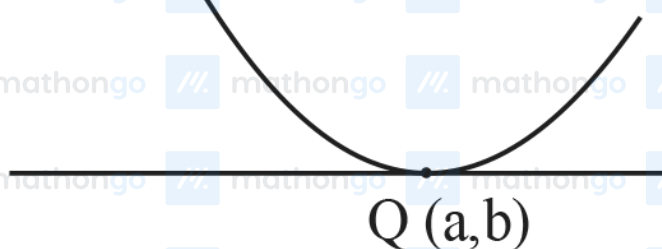
$$y'_P = 22$$

$\therefore$ tangent to curve at P

$$y + 1 = 22(x - 2)$$

$$y = 22x - 45$$

$$y = x^3 - x^2 + x$$



$$\left. \frac{dy}{dx} \right|_{C_2} = 3x^2 - 2x + 1$$

$$\left. \frac{dy}{dx} \right|_Q = 3a^2 - 2a + 1$$

$$\text{Hence } 3a^2 - 2a + 1 = 22$$

$$\therefore 3a^2 - 2a - 21 = 0$$

$$3a^2 - 9a + 7a - 21 = 0$$

$$(3a + 7)(a - 3) = 0$$

$$a = 3$$

$$a = -7/3$$

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$$\text{from curve } b = a^3 - a^2 + a$$

$$a = 3$$

$$b = 21 \quad |2a + 9b| = 195$$

at $a = -7/3$ tangent will be parallel

Hence it is rejected

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